

UMESH SUBBA LIMBU

Staff Software Engineer

Kathmandu, Nepal | umes4ever@gmail.com | [LinkedIn](#) | [Portfolio](#)

Professional Summary

Staff Software Engineer with 10+ years of experience designing, building, and scaling distributed systems and AI-native platforms. Specialized in LLM-based agentic workflows, advanced RAG architectures, and production engineering. Proven track record of leading cross-functional teams of 10–15 engineers, owning system design end-to-end, and delivering practical improvements in system reliability, latency, operational costs, and developer workflows.

Technical Core Competencies

AI & LLM Systems	Agentic Workflows, Advanced RAG, pgvector, Claude code, LiteLLM, Voyage AI Embeddings, Prompt Engineering, Context Optimization
Languages	Python, Java, TypeScript, Go (basic)
Data & Backend	Spring Boot, NestJS, Next.js, PostgreSQL, MongoDB, Redis, Elasticsearch, REST APIs, Microservices, Event-Driven Architecture
Cloud & DevOps	AWS (RDS, SQS, Lambda, S3, Cognito, ECR, CodePipeline), Docker, Kubernetes, CI/CD
Observability & QA	OpenTelemetry, Prometheus, Grafana, Playwright, JUnit, Mockito, Integration Testing

Professional Experience

Nov 2025 – Present **Staff Software Engineer**
Fusemachines Nepal Pvt. Ltd. • Kathmandu, Nepal

- Designed and built a vector memory layer using pgvector, improving semantic retrieval and reducing redundant LLM context token usage by 25–30%.
- Built a reliable distributed workflow orchestration layer using Python Procrastinate to handle

thousands of async jobs daily with built-in retries, failure recovery, and logging.

- Integrated Playwright end-to-end testing into our CI/CD pipelines, reducing production regressions by 35% and making releases much more predictable.
- Reduced critical path latency across AI workflows by 20% through contextual caching, retrieval optimization, and prompt refactoring.

Jan 2024 – Oct 2025 Assistant Technology Manager / Project Lead

Fusemachines Nepal Pvt. Ltd. • Kathmandu, Nepal

- Led cross-functional teams of 10–15 engineers, taking multiple enterprise platforms from initial design all the way through to live production.
- Designed scalable, HIPAA/GDPR-compliant backend systems to ensure secure and auditable handling of regulated enterprise data.
- Reduced deployment-related production incidents by 25–30% by improving CI/CD health checks, code review guidelines, and release governance.
- Managed system design reviews for distributed microservices, making core decisions on service boundaries, asynchronous communication, and data schemas.
- Lowered AWS infrastructure costs by 15% through infrastructure right-sizing, compute instance scheduling, and database query optimization.

Jan 2022 – Dec 2023 Assistant Technology Manager

Fusemachines Nepal Pvt. Ltd. • Kathmandu, Nepal

- Managed technical roadmaps to scale multiple internal products from early MVPs into stable production systems serving thousands of active users.
- Improved engineering delivery efficiency by 15–20% by identifying cross-team dependencies early and removing development blockers.
- Introduced a standardized evaluation framework to vet backend and AI tools, ensuring consistent technology adoption across engineering teams.

Jan 2021 – Dec 2021 Senior Software Engineer / Team Lead

Fusemachines Nepal Pvt. Ltd. • Kathmandu, Nepal

- Supervised a team of 10+ software engineers, delivering over 30 stable production releases a year.
- Resolved over 200 production issues and reduced recurring bugs by ~30% by introducing structured triage processes and better system observability.

Jan 2020 – Dec 2020 Senior Software Engineer

Fusemachines Nepal Pvt. Ltd. • Kathmandu, Nepal

- Built and maintained core backend services using Spring Boot and MongoDB, handling over 100k daily multi-tenant REST API requests.
- Improved critical system path throughput by 15% via database indexing, schema updates, and query optimization.
- Mentored junior engineers and ran internal technical training sessions.

Jan 2019 – Dec 2019 Software Engineer

Fusemachines Nepal Pvt. Ltd. • Kathmandu, Nepal

- Developed backend APIs and core microservices using Spring Boot for applications running under live user traffic.
- Contributed directly to database modeling, system design reviews, and general performance tuning.

Education & Credentials

May 2015 Bachelor of Engineering (B.E.) in Computer Science | NMAMIT, India (CGPA: 9.42/10.0)

Nov 2025 [AWS Certified Cloud Practitioner](#) | Amazon Web Services

Feb 2021 M320: Data Modeling | MongoDB University